SNOWFLAKE DATA ENGINEERING

From Fundamentals to Production-Ready Data Pipelines

From Snowflake fundamentals to reliable, production-oriented data pipelines.

Modern data engineering is not just about loading tables. It means making deliberate decisions about architecture, data quality, performance, security, reliability, and cost. This practical guide shows how those decisions connect in the Snowflake data platform.

Progress from Snowflake architecture and SQL to ingestion, transformation, orchestration, analytics, and AI. Each chapter connects concepts with engineering trade-offs, guided exercises, common failure modes, and companion GitHub resources.

WHAT YOU WILL LEARN

- Design and implement robust ETL and ELT data pipelines.
- Master practical SQL and Python for data processing.
- Understand data warehouses, data lakes, and lakehouse architectures.
- Explore batch processing, streaming, and orchestration (Snowpipe, Streams, Tasks, Dynamic Tables).
- Apply data quality, testing, observability, and governance practices.
- Build cloud-based data solutions and production-oriented workflows.

FROM THEORY TO PRACTICE

Explore practical engineering exercises, real-world case studies, architectural examples, and companion GitHub resources designed to connect technical concepts with hands-on implementation.

WHO SHOULD READ THIS BOOK?

Designed for aspiring and practicing Data Engineers, Analytics Engineers, Software Engineers, Data Analysts, and university students seeking a structured understanding of modern data engineering with Snowflake.



ABOUT THE AUTHOR

Arman Golbidi is a Data Engineer with professional experience in Azure, Python, SQL, Power BI, and data analytics. He holds an MSc in Data-Centric Engineering from LUT University, Finland. His background combines practical data platform development, analytics engineering, and applied machine learning.

COMPANION RESOURCES

Source code • Practical exercises • GitHub projects

Access the companion repository for code, datasets, and additional learning materials.

SCAN FOR PROJECT REPOSITORY

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